

Covariate-Augmented TimesFM Hindcast Workflow

Hourly monitoring records are corrected, decomposed into covariates and residual structure, and tested against a regional earthquake catalogue

1 DATA AND COVARIATE PREPARATION

Hourly observations

Water level, pressure and rainfall, 2013-2025

Quality control

Step and polarity correction; gap flagging

Daily aggregation

Daily means for water level and pressure

Thirteen covariates

Pressure lags, rainfall accumulations, trend, harmonics

Rolling protocol

365-day context | 30-day horizon | 7-day stride | 622 windows

2 TWO-STAGE FORECASTING

Ridge regression

Linear covariate fit
 $\alpha = 1.0$

Residual extraction

$$r(t) = y(t) - \hat{y}_{cov}(t)$$

TimesFM 2.5

Residual forecast, quantiles q10-q90

Recombination

$$\hat{y}_{final} = \hat{y}_{cov} + \hat{r}$$

Retrospective hindcast

Observed pressure and rainfall are used over each forecast horizon.
Operational use would require separate covariate forecasts.

3 ANOMALY INTERPRETATION AND EARTHQUAKE ASSOCIATION

Weighted score

$$S(t) = \sum w_i d_{it} \sum w_i \\ w_i = 1/h_i$$

Calibration thresholds

Expanding monthly T90 / T95 from strictly earlier scores

Cause classification

Rainfall, pressure, mixed or residual screening

Earthquake association

Sequence grouping, permutation, FDR, Molchan

Outputs: anomaly catalogue | cause classes | timing association statistics